



THE KENYA NATIONAL EXAMINATIONS COUNCIL
Kenya Certificate of Secondary Education



443/2 –

AGRICULTURE
(THEORY)

– Paper 2

Nov. 2017 – 2 hours

Name Index Number

Candidate's Signature Date

Instructions to candidates

- (a) Write your name and index number in the spaces provided above.
- (b) Sign and write the date of examination in the spaces provided above.
- (c) This paper consists of **three** sections: **A**, **B** and **C**.
- (d) Answer **all** the questions in sections **A** and **B**.
- (e) Answer any **two** questions in section **C**.
- (f) All answers should be written in the spaces provided in the question paper.
- (g) This paper consists of **16** printed pages.
- (h) Candidates should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.
- (i) Candidates should answer all the questions in English.

For Examiner's Use Only

Section	Questions	Maximum Score	Candidate's Score
A	1–16	30	
B	17–20	20	
C		20	
		20	
Total Score		90	



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Turn over



SECTION A (30 marks)

*Answer **all** the questions in this section in the spaces provided.*

1. Name **one** breed of livestock under each of the following categories:
 - (a) dairy cattle (½ mark)
 - (b) dual purpose cattle (½ mark)
 - (c) dual purpose sheep (½ mark)
 - (d) hair goat (½ mark)

2. State **four** egg abnormalities observed during egg candling. (2 marks)
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3. Give **four** reasons for livestock breeding. (2 marks)
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4. Give **two** reasons for hoof trimming. (1 mark)
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5. State **four** signs of Anthrax disease. (2 marks)

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6. Name **four** methods of dehorning. (2 marks)

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7. Give **four** reasons for keeping dairy animals healthy. (2 marks)

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8. State **four** features of a clean milkman. (2 marks)

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9. Name **four** maintenance practices for a wire fence. (2 marks)

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10. State **four** reasons why dorper breed of sheep is good for mutton production. (2 marks)

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11. State **four** control measures for tsetse flies. (2 marks)

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12. Explain how each of the following is measured in cattle:

(a) blood temperature (½ mark)

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(b) respiratory rate (½ mark)

.....

(c) pulse rate. (½ mark)

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13. State the gestation period for each of the following:

(a) cow (½ mark)

.....

(b) sow (½ mark)

.....

(c) ewe (½ mark)

.....

(d) rabbit. (½ mark)

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14. State four features of large white breed of pigs. (2 marks)

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15. State four functions of worker bees in a bee colony. (2 marks)

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16. State three ways of caponisation in poultry. (1½ marks)

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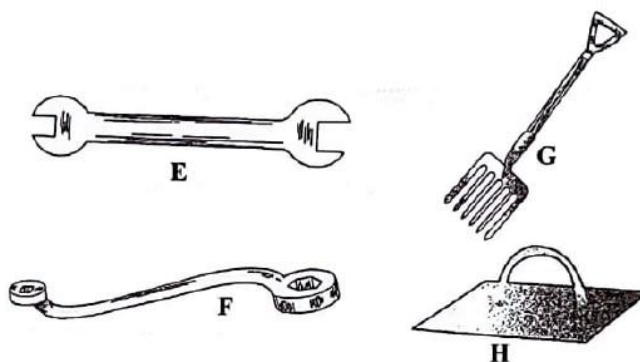
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SECTION B (20 marks)

Answer all the questions in this section in the spaces provided.

17. The diagrams below represent farm tools and equipment.



- (a) Identify the tools labelled **E** and **F**.

E (1 mark)

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F (1 mark)

.....

- (b) State **one** use of each of the tools labelled **F** and **G**

F (1 mark)

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G (1 mark)

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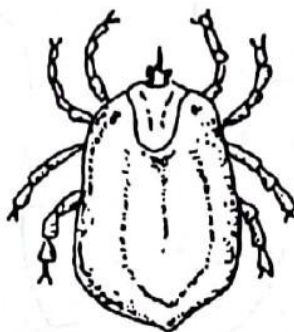
- (c) State **one** advantage the tool labelled **F** has over **E**. (1 mark)

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18. The diagram below illustrates an external parasite of livestock.



- (a) Identify the parasite. (1 mark)

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- (b) Apart from eggs and adults, name **two** other developmental stages of the parasite illustrated above. (1 mark)

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- (c) Name **two** livestock diseases transmitted by the illustrated parasite. (1 mark)

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- (d) Explain **two** ways in which the environment is altered to control the parasite illustrated above. (2 marks)

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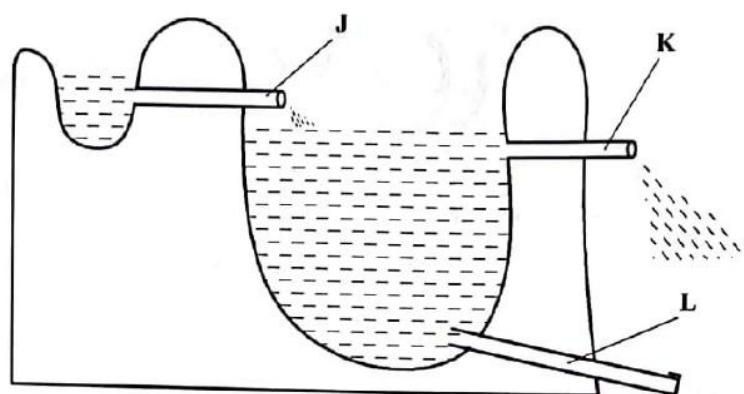
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19. The diagram below shows a cross section of a fish pond



(a) Name the part labelled

J

(1 mark)

K

(1 mark)

(b) State one function of the part labelled L

(1 mark)

(c) Give one reason why a pond should have a deep end.

(1 mark)

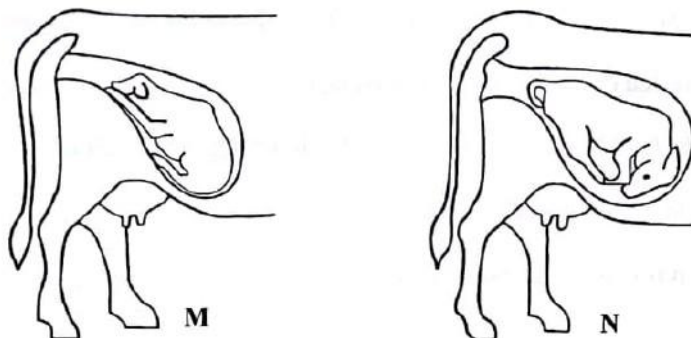
(d) Explain how fingerlings are introduced into a pond?

(1 mark)





20. The diagram below shows calf presentations during parturition.



(a) Name the presentation labelled

M (1 mark)

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N (1 mark)

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(b) Which **one** of the two presentations requires the services of a qualified stockman? (1 mark)

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(c) State **two** signs of parturition observed on the vulva. (2 marks)

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SECTION C (40 marks)

Answer any two questions from this section in the spaces provided after question 23.

- 21.** (a) Describe **ten** characteristics of a good layer. (10 marks)
- (b) Describe the milking procedure under the following sub-headings:
- (i) pre-milking practices (5 marks)
- (ii) procedure of proper milking. (5 marks)
- 22.** (a) Describe Brucellosis disease in cattle under the following sub-headings:
- (i) causal organism (1 mark)
- (ii) signs of attack (4 marks)
- (iii) control measures. (5 marks)
- (b) Explain **five** factors that affect the digestibility of a feed material. (5 marks)
- (c) Explain **five** requirements of an artificial brooder in poultry rearing. (5 marks)
- 23.** (a) State **five** harmful effects of lice in pigs. (5 marks)
- (b) Explain **five** structural requirements for a grain store. (5 marks)
- (c) Describe **five** maintenance practices for the water cooling system of a tractor. (5 marks)
- (d) Give **five** differences between diesel and petrol engines. (5 marks)

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